



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/825158

Date:

07/02/2023(12:15:28)

Subject: Approval of New LMI “GUIDELINES TO CONTROL STEAM PURITY IN BOILERS AFTER OVERHAUL/MODIFICATION/BOILER COLD START UP”

REF.: SMPP/O&M/MM/PP/

Date: 07.02.2023

Subject: Approval of New LMI “GUIDELINES TO CONTROL STEAM PURITY IN BOILERS AFTER OVERHAUL/MODIFICATION/BOILER COLD START UP”

LMI No: COS-ISO-00-OGN/OPS/SYST/056

This is a new LMI which was prepared as per the COS-ISO-00-OGN/OPS/SYST/056 guidelines.

Put up for approval please

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A new LMI " Guidelines to control steam purty in Boilers after Overhaul/Modification/Boiler Cold start up" has been prepared based on OGN with ref no. OGN-OPS-SYST-056 Rev-0.

Henceforth the relevant protocols/records shall be maintained/followed as per details shared in this LMI.

CA may please provide approval for this new LMI to implement the guidlines to control steam purity in Boilers.

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/SYST/056

ISSUE NO.0; REVISION NO. 0; DATE: FEB'2023

**GUIDELINES TO CONTROL STEAM
PURITY IN BOILERS AFTER
OVERHAUL/MODIFICATION
/BOILER COLD START UP**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	GUIDELINES TO CONTROL STEAM PURITY IN BOILERS AFTER OVERHUAL /MODIFICATION /BOILER COLD START UP
<u>2.a) DOCUMENT CODE</u> b) REVISION/ISSUE NO.&DATE	LMI/OGN/OPS/SYST/056 <u>ISSUE-0; REV.-0; FEB'2023</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>COS-ISO-00-OGN/OPS/SYST/056</u>
<u>4.PREPARED BY: SENIOR MANAGER</u> (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY:</u> DGM(MM-BMD) AGM (BMD) AGM(OPN) AGM(CHEM) GM (MTC.) GM (O&M)	SIGNATURE----- SIGNATURE_____ SIGNATURE_____ SIGNATURE_____ SIGNATURE_____
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE:----- HOP (SIMHADRI)
<u>7. DATE OF ISSUE</u>	
<u>8. INDEX LIST</u>	ENCLOSED.
<u>9. DISTRIBUTION LIST</u>	ENCLOSED.

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2.0 INTRODUCTION

The Objective of this Guidelines is for achieving steam purity by removing condensate, exfoliated scales, weld slags, purge paper and any loose debris before turbine entry in units coming after boiler overhaul/modification and Boiler cold start up (boiler cold start up is when PTSH/DIV/FSH coil any available acromat temperature < 110°C, where acromats are not available flue gas temperature at APH inlet are less than 110°C).

The objective of this LMI is to establish a system for controlling steam purity in boilers after overhaul/modification/boiler cold start up.

3.0 SCOPE

The scope of this LMI is supercritical boilers, sub critical drum and sub critical once through boilers .

4.0 REFERENCE DOCUMENT

Operation information note COS-ISO-00-OGN/OPS/SYST/056 circulated by operation services-NTPC corporate center.

5.0 SUPERSEDED DOCUMENT

Not applicable

6.0 SPECIFIC INSTRUCTIONS

The objective of this LMI is sought to be achieved by

Boiler water filling quality (Refer Chemistry OGN/OPS/CHEM/017):

Boiler filling to be done with “flow swing” (three cycle with each cycle of 10% BMCR of feed water flow for thirty minutes then keeping no flow (0 T/Hr.) for 10 min) keeping drain open for minimum 15 minutes after water starts coming from Drain.

Sample Name: Eco Inlet/ Deaerator outlet	
Parameters	Limiting Values
pH	9.2-9.6
Silica (ppb), max	30
Turbidity (NTU), max	0.7
Total iron (dissolved and suspended), ppb, max	50
Crud(ppb), Max	50
DO (ppb), max	100

b) **Boiler water light up quality (Refer Chemistry OGN/OPS/CHEM/017)**

Sample name: Eco inlet	
Parameters	Limiting Values
pH	9.2-9.6
Silica, ppb, max	30
Total Iron (dissolved& suspended) ppb, max	50
Turbidity (NTU), max	0.7
Crud, ppb, max	50
Cation Conductivity, $\mu\text{S}/\text{cm}$, max	0.5
DO, ppb max	100

7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS

Cleaning of tubes and headers before being put to use

1. Prevention of ingress of foreign material inside tube and headers while storage by keeping tubes, coils, and headers on stand during storage .All openings to be covered by outside metallic caps. All caps should be either self-locking type/ tight fit or to be fixed with brown tape to avoid fall off during storm/rain/ tube handing.
2. Cleaning of tubes and headers before being used by sponge ball test .After sponge ball test, tubes to be steam blown by pressure 12-16 Ksc and then dried by air and to be recapped immediately before transportation to site.
3. Headers and stubs to be checked for any foreign material visually.
4. New tubes to be checked for any plastic cap remains inside tube.

RESPONSIBILITY: HOD (BMD)

Prevention of ingress of material during Maintenance work

- 1.Any loose material lying near work which can fall inside tubes to be removed.
- 2.All tube / coil ends must be kept capped and during edge preparation only tubes in which work is being done to be uncapped and to be recapped after edge preparation.
- 3.Caps used must be self-locking /tight fit/fixed by brown tape.
- 4.Water soluble purge paper dam to be used during welding in coil circuit.

RESPONSIBILITY: HOD (BMD)

Operation of Drain and Vent line

- Each Drain line valves to be closed only after ensuring that there is no choking in line and drain is through for each drain line (observing flow noise from line and thermocouple to be fixed on critical lines like second pass C header drains etc.)
- Vent line valves also to be closed only after ensuring that there is no choking in line (Continuous fluid oozing from line).
- Combined protocol for ensuring that there is no choking of vents and drains with operation is to be carried out before box up.

Resp: HOD (OPN)

Flushing after Hydro

- After hydro test, boiler to be flushed from drain header and each drain line valve to be reopened after hydro test.

Hot Draining after first Boiler Light Up

- Boiler to be hot drained after reaching pressure of 10 KSC for minimum 15 minutes and before opening MS stop valve (Fire not to be killed) and all the vents and lowest point drain lines to be kept open while hot draining.

Opening ERVs/ EBV/ Hydraulic Safety Valve at Lower Pressure-

MS ERVs/ EBV/ Hydraulic Safety Valve (02 separate valves) to be operated from control room once for 05 min each path (LHS & RHS) at pressure 10-15 KSC before opening MS Stop Valves.

Hot Clean Up Cycle/ Steam Dumping (Refer Chemistry OGN/OPS/CHEM/017)

- Hot clean-up cycle starts after boiler light up and feed water temperature shall be maintained as per OEM recommendation during hot clean up cycle.
- Start boiler condensate transfer pump to transfer FTDT water to the condenser. frequent monitoring of SST/ FTDT water is mandatory to analyze the silica & turbidity. If any parameter abruptly increases, then stop boiler condensate transfer pump and start dumping to atmosphere till parameters are in limit to pass through CPU. CPU charging should be done judiciously after taking proper care of Differential Pressure, to avoid any damage due to crud.
- All condensate flow must pass through CPU only. Effective clean-up cycle requires around 30% TMCR condensate flow.
Note: LTSH I/L & MS samples shall be charged in SWAS for monitoring of online chemical parameters. Sp. conductivity & ACC should be charged. Chiller unit shall be made ready before sample lines are charged.
- After charging of HP & LP bypass system, if CPU inlet water quality is beyond the acceptable limit, then start 50-100 % condensate dumping to atmosphere. During this period change DM makeup to deaerator and stop hotwell to deaerator flow. LP dosing to be done at suction of running MD/TD-BFPs only.
- In case of frequent choking of CEP Strainers, stop the dumping process and hotwell needs to be inspected and cleaned. Keeping Reheater vent line open after Charging of HP Bypass.
- Before increasing firing, reheater vent line to be kept open till steam starts coming with pressure (local check – Must in case of hydro test of reheater) and turbine side Strainer drain lines to be kept open.

Extended Steam Blowing from ERVs/ EBV/ Hydraulic Safety Valves /MS Blow Offline from Control Room

- Steam blowing of Superheater coils, and Reheater coils after MS Pr is 30 -60 KSC first by opening ERV/ EBV/ Hydraulic Safety Valve of SH and RH each path (Only 02 separate valves not all valves) for 03 times for minimum 05 minutes. Subsequent blowing to be decided based on steam purity parameters.
- HP /LP bypass should not be used for blowing off or reducing pressure as it can cause sharp saturation temperature drop and this can cause exfoliation inside SH &RH tubes. Rate of increase of fuel firing to be controlled such that superheater tube metal temperature rise are $< 3^{\circ}\text{C}/\text{min}$ and there are no abrupt rise in sprays. Rate of firing to be reduced in case of any sharp increase in MTM and acromats.
- In case of boiler Box up because of any reason pressure to be reduced by operating ERVs for maximum 5 min for each opening.

Note: HP Bypass should not be opened after boiler shut down to depressurize the boiler or to check reheater leakages. Reheater air pressurization (instrument line provided for preservation to be used for reheater air pressurization check)

Resp: – HOD(OPN/CHEM)

8. EXCEPTION REPORTING

The exceptions encountered during inspection are to be reported as per the following format
----- Mech. Mtc.(Pr.Parts)

Description of exception	Unit No.	Location of exception	Remarks

9.0 RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to control steam purity in boilers after overhaul/modification/boiler cold start up as detailed in the LMI is with HOS (Pressure Parts),HOS (Chemistry) and HOS(Operation).

10. RECORD KEEPING

The records to be maintained under this LMI include

- a) Compliance record of the survey and mtc. activities detailed under this LMI
- b) Exception reporting

The above records are to be maintained by Area Engr. Pr. Parts Group.

11.REVIEW

This document shall be reviewed by HOP (SIMHADRI) on TWO yearly basis.

12.0 REVISION INDEX

(for changes/revision)

(for changes/revision)

[illegible]

13.0 DISTRIBUTION LIST

[illegible]

<u>Checklist/Protocol: Boiler Water Filling Quality</u>			
Unit		Date:	
Sample Name: Eco Inlet/ Deaerator outlet			
S.No	Parameters	Limiting Values	Achieved Values
1	pH	9.2-9.6	
2	Silica (ppb), max	30	
3	Turbidity (NTU), max	0.7	
4	Total iron (dissolved and suspended), ppb, max	50	
5	Crud(ppb), Max	50	
6	DO (ppb), max	100	

CHEMISTRY

OPERATION

BMD

<u>Checklist/Protocol: Boiler Water Light Up Quality</u>			
Unit		Date:	
Sample Name: Eco Inlet			
S.No	Parameters	Limiting Values	Achieved Values
1	pH	9.2-9.6	
2	Silica, ppb, max	30	
3	Total Iron (dissolved& suspended) ppb, max	50	
4	Turbidity (NTU), max	0.7	
5	Crud, ppb, max	50	
6	Cation Conductivity, μS/cm, max	0.5	
7	DO, ppb max	100	

CHEMISTRY

OPERATION

BMD

Checklist/Protocol: Hot draining of Boiler after Light up (Applicable after Boiler Overhaul/Modification)			
Unit		Date:	
Separator Pressure to be achieved: 10-15 Ksc		Duration	15 minutes
S.No	Work Description	Status	Remarks
1	Bottom Ring Header drains		
2	Intermediate Header Drains		
3	Roof Inlet Headers Drains		
4	2 nd Pass ring header Drains		
5	Eco Drains		
6	MS Line Drains (Before MS stop Valve)		

BMD

OPERATION

Checklist/Protocol: MS & RH ERVs / EBV/ Hydraulic Safety Valves Operation / MS Blow off line valves (Applicable for Boiler Cold startup/Overhaul/Modification)								
Unit			Date:					
(A) ERVs/blow off line valves Operation at Lower Pressure (Before Opening MS Stop Valves)								
S.no	Description	At MS Pressure (10-15 ksc)			Duration (2 min.)			Remarks
1	MS ERV/ EBV/ Hydraulic Safety Valves LHS & RHS							
(B) ERVs/blow off line valves operation at Higher Pressure								
S.no	Description	At MS Pressure (50-60 ksc)			Duration (5 min.)			Remarks
		Cycle			Cycle			
		1	2	3	1	2	3	
1	MS ERV/ EBV/ Hydraulic Safety Valves/blow off line valves LHS & RHS							
		At RH Pressure (10-15 Ksc)			Duration (5 min.)			Remarks
		Cycle			Cycle			
		1	2	3	1	2	3	
2	RH ERV/ EBV/ Hydraulic Safety Valves LHS- 1 no. & RHS- 1 no.							

Note: LHS & RHS ERVs/ EBV/ Hydraulic Safety Valves/blow off line valves to be operated at same time.

BMD

OPERATION

CHECKLIST/ PROTOCOL

Checklist/Protocol: MS & RH ERVs/ EBV/ Hydraulic Safety Valves Operation during Boiler Box up

Unit			Date:		
S.no	Description	At MS Pressure (50-60 Ksc)	Duration (5 min.)		Remarks
1	MS ERV LHS & RHS				
		At RH Pressure (10-15 Ksc)	Duration (5 min.)		Remarks
2	RH ERV LHS- 1 no. & RHS- 1 no.				

Note: LHS & RHS ERVs/ EBV/ Hydraulic Safety Valves to be operated at same time.

BMD

OPERATION